Supplementary Materials: Critical-Damping Control in Viral Dynamics

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Contents

1 S1. χ Estimation Methods and Validation

1.1 S1.1. Autoregressive Estimation: Full Derivation

For sparse viral load time series $\{V(t_i)\}$ with $i = 1, \ldots, n$, the autoregressive approach models log-transformed measurements $x_i = \log_{10} V(t_i)$ as a discrete-time stochastic process.

Model specification. An AR(p) model with order p takes the form:

$$x_n = \sum_{k=1}^{p} a_k x_{n-k} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

The characteristic polynomial is:

$$\Phi(z) = 1 - \sum_{k=1}^{p} a_k z^{-k} = 0. \tag{2}$$

For viral dynamics with underlying damped oscillatory behavior, the dominant roots are a complex conjugate pair $z_\pm = re^{\pm i\theta}$ where $r < 1$ (stability) and $\theta \in (0, \pi)$ (oscillation frequency).

Continuous-time mapping. The discrete-time eigenvalues $z_\pm$ relate to continuous-time eigenvalues $\lambda_\pm = -\gamma \pm i\omega$ through:

$$z_\pm = e^{\lambda_\pm \Delta t} = e^{-\gamma \Delta t} e^{\pm i\omega \Delta t}, \tag{3}$$

where Δt is the mean sampling interval. Taking magnitude and argument:

$$r = |z_\pm| = e^{-\gamma \Delta t} \quad \Rightarrow \quad \gamma = -\frac{\ln r}{\Delta t}, \tag{4}$$

$$\theta = \arg(z_\pm) = \omega \Delta t \quad \Rightarrow \quad \omega = \frac{\theta}{\Delta t}. \tag{5}$$

The stability ratio follows:

$$\chi = \frac{\gamma}{2|\omega|} = -\frac{\ln r}{2\theta}. \tag{6}$$

Order selection. Model order p is chosen via Akaike Information Criterion:

$$\mathrm{AIC}(p) = n \ln \hat{\sigma}_p^2 + 2p, \tag{7}$$

where $\hat{\sigma}_p^2$ is the residual variance for order p. Optimal p^* minimizes AIC. For viral dynamics, typically $p^* \in \{2, 3\}$.

1.2 S1.2. Handling Irregular Sampling

When sampling intervals vary, a state-space formulation with Kalman filtering accommodates irregular time stamps. The observation equation becomes:

$$x(t_i) = \mathbf{H}\mathbf{z}(t_i) + v_i, \tag{8}$$

where $\mathbf{z}(t)$ evolves according to the continuous-time state equation and $\mathbf{H}$ is the observation matrix. The Kalman filter update step at each t_i accounts for the actual elapsed time $\Delta t_i = t_i - t_{i-1}$.

1.3 S1.3. Censored Data and Tobit Likelihood

Viral load assays have a lower limit of detection (LOD). Observations below LOD are right-censored. The Tobit likelihood is:

$$\mathcal{L}(\mathbf{a}, \sigma^2 | \mathbf{x}) = \prod_{i \in \mathcal{O}} \phi(x_i | \mu_i, \sigma^2) \prod_{j \in \mathcal{C}} \Phi\left(\frac{x_{\text{LOD}} - \mu_j}{\sigma}\right), \tag{9}$$

where $\mathcal{O}$ is the set of observed (above LOD) indices, $\mathcal{C}$ is the set of censored indices, ϕ is the Gaussian PDF, Φ is the Gaussian CDF, and $\mu_i = \sum_{k=1}^{p} a_k x_{i-k}$ is the AR prediction.

Maximum likelihood estimation proceeds via the EM algorithm:

- E-step: Impute censored values using truncated normal conditional expectations.
- M-step: Update AR coefficients $\{a_k\}$ and variance σ^2 via least squares on augmented data.

Convergence is assessed via log-likelihood change $< 10^{-4}$.

1.4 S1.4. Monte Carlo Validation

Synthetic data generation. Generate $n_{\text{sim}} = 1000$ synthetic viral load series with known $\chi_{\text{true}} \in [0.5, 1.5]$. For each series:

1. Draw $\gamma_{\text{true}} \sim \mathcal{U}(0.05, 0.3)$ day^{-1}, $\omega_{\text{true}} \sim \mathcal{U}(0.05, 0.2)$ day^{-1}.
2. Compute $\chi_{\text{true}} = \gamma_{\text{true}}/(2\omega_{\text{true}})$.
3. Generate time series: $x(t) = x_0 e^{-\gamma_{\text{true}} t} \cos(\omega_{\text{true}} t) + \varepsilon(t)$, where $\varepsilon(t) \sim \mathcal{N}(0, \sigma^2_{\text{noise}})$ with coefficient of variation CV = 20%.
4. Sample at irregular intervals: $t_i \sim t_{i-1} + \mathcal{U}(2, 7)$ days, yielding $n_{\text{obs}} \in [12, 25]$ points.
5. Apply AR(2) estimation to recover $\hat{\chi}$.

Performance metrics.

- Relative error: $e_{\text{rel}} = |\hat{\chi} - \chi_{\text{true}}|/\chi_{\text{true}}$.
- Bias: bias $= \mathbb{E}[\hat{\chi} - \chi_{\text{true}}]$.
- RMSE: RMSE $= \sqrt{\mathbb{E}[(\hat{\chi} - \chi_{\text{true}})^2]}$.
- Coverage: Proportion of 95% bootstrap CIs containing χ_{true}.

Results summary.

χ Range	Mean e_{rel}	Bias	RMSE	Coverage	n median
0.5–0.7	9.2%	−0.02	0.07	94.1%	18
0.7–1.0	7.8%	+0.01	0.06	95.3%	17
1.0–1.5	8.9%	−0.03	0.09	93.8%	19
Overall	**8.3%**	−0.01	0.07	94.4%	18

Table 1: AR(2) estimator performance across χ regimes (1000 synthetic series).

The estimator is approximately unbiased with coverage near nominal 95%. Performance degrades when $n < 12$ or CV $> 30\%$.

Order	Mean e_{rel}	RMSE	AIC_{avg}	Overfit %
AR(1)	14.2%	0.12	52.3	0%
AR(2)	8.3%	0.07	48.1	3.2%
AR(3)	8.7%	0.08	49.5	12.8%
AR(4)	9.5%	0.09	51.2	24.5%

Table 2: Performance vs AR order. AR(2) minimizes error while avoiding overfitting.

1.5 S1.5. Sensitivity to Model Order

1.6 S1.6. Code Availability

Python implementation available at `github.com/SymCUniverse/viral-chi/estimation`. Key functions:

- `ar_estimate_chi(x, t, order=2)`: Main estimator with automatic order selection.
- `bootstrap_ci(x, t, n_boot=5000)`: Bootstrap confidence intervals.
- `tobit_fit(x, t, lod)`: Censored data handling.

2 S2. Nonlinear Validation and Model Extensions

2.1 S2.1. Nonlinear Amplitude Corrections

For large excursions $|X - X^*|/X^* > 2$, the effective damping ratio becomes amplitude-dependent. A perturbation expansion yields:

$$\chi_{\text{eff}}(A) \approx \chi_0 + \alpha_2 A^2 + \mathcal{O}(A^4), \tag{10}$$

where A is the oscillation amplitude and α_2 depends on the nonlinearity structure (logistic growth term).

Numerical integration of the full nonlinear system shows χ_{eff} remains within ± 0.15 of linear prediction for $A < 2X^*$, validating the linearization regime statement in the main text.

2.2 S2.2. Multi-Strain Resistance Model

Extend to drug-sensitive (X_S) and resistant (X_R) populations:

$$\dot{X}_S = r_S X_S \left(1 - \frac{X_S + X_R}{K}\right) - \alpha X_S Y - u(t) X_S, \tag{11}$$

$$\dot{X}_R = r_R X_R \left(1 - \frac{X_S + X_R}{K}\right) - \alpha X_R Y, \tag{12}$$

$$\dot{Y} = -\delta Y + \beta(X_S + X_R). \tag{13}$$

The 3×3 Jacobian has one real eigenvalue and a complex conjugate pair. The effective χ is computed from the complex pair. As resistance emerges (X_R/X_S increases), χ shifts due to changing replication rate $r_{\text{eff}} = (r_S X_S + r_R X_R)/(X_S + X_R)$, typically increasing if $r_R > r_S$.

2.3 S2.3. Spatial Compartmentalization

Three-compartment model: blood (B), lymph nodes (L), CNS (C).

$$\dot{X}_B = rX_B - \alpha X_B Y_B - u_B(t)X_B + k_{LB}X_L - k_{BL}X_B, \tag{14}$$
$$\dot{X}_L = rX_L - \alpha X_L Y_L - u_L(t)X_L + k_{BL}X_B - k_{LB}X_L, \tag{15}$$
$$\dot{X}_C = rX_C - \alpha X_C Y_C - u_C(t)X_C + k_{BC}X_B - k_{CB}X_C, \tag{16}$$
$$\vdots \text{(control equations)} \tag{17}$$

The 6×6 Jacobian has three complex conjugate pairs, one per compartment. The slowest mode (smallest $|\lambda|$) dominates long-term dynamics; this is typically the CNS due to poor drug penetration ($u_C \ll u_B$).

2.4 S2.4. Innate/Adaptive Immune Disaggregation

Separate $Y(t)$ into innate (Y_I, interferon/NK) and adaptive (Y_A, CTL/antibody):

$$\dot{Y}_I = -\delta_I Y_I + \beta_I X, \tag{18}$$
$$\dot{Y}_A = -\delta_A Y_A + \beta_A X Y_I. \tag{19}$$

This introduces two timescales: $\tau_I = 1/\delta_I \sim$ 1–2 days (innate), $\tau_A = 1/\delta_A \sim$ 7–14 days (adaptive). The dominant χ corresponds to the adaptive timescale in chronic infections, innate timescale in acute infections.

3 S3. HIV-1 Simulation Details

3.1 S3.1. Complete Model Equations

Target cell (T), infected cell (I), virus (V), CTL (E) model:

$$\dot{T} = \lambda - dT - (1-\eta)\beta TV, \tag{20}$$
$$\dot{I} = (1-\eta)\beta TV - \delta I - kEI, \tag{21}$$
$$\dot{V} = pI - cV, \tag{22}$$
$$\dot{E} = -\delta_E E + \beta_E IE, \tag{23}$$

where $\eta = u(t)$ is drug efficacy (protease inhibitor), $\lambda = 10^4$ cells/(μL·day) is thymic production, $d = 0.01$ day^{-1} is T cell death, $\beta = 2.4 \times 10^{-5}$ (μL/virion·day) is infectivity, $\delta = 1.0$ day^{-1} is infected cell death, $k = 10^{-4}$ (μL/cell·day) is CTL killing, $p = 2000$ virions/cell is production, $c = 23$ day^{-1} is viral clearance, $\delta_E = 0.1$ day^{-1} is CTL death, $\beta_E = 0.01$ (μL/cell·day) is CTL stimulation.

3.2 S3.2. Parameter Sources

3.3 S3.3. Dose-Response Curve

Drug efficacy $\eta(C)$ follows Hill function:

$$\eta(C) = \frac{E_{\max} C^n}{EC_{50}^n + C^n}, \tag{24}$$

with $E_{\max} = 0.95$, $n = 1.5$, $EC_{50} = 100$ ng/mL. For $u(t) \in [0,1]$ dimensionless efficacy, we set $u = \eta(C)$ with inverse mapping.

Parameter	Value	Units	Source
λ	10^4	cells/(μL·day)	Ho 1995
d	0.01	day^{-1}	Perelson 1999
β	2.4×10^{-5}	μL/(virion·day)	Nowak 2000
δ	1.0	day^{-1}	Perelson 1997
k	10^{-4}	μL/(cell·day)	Wodarz 1999
p	2000	virions/cell	Perelson 1996
c	23	day^{-1}	Ramratnam 1999
δ_E	0.1	day^{-1}	De Boer 2003
β_E	0.01	μL/(cell·day)	Estimated

Table 3: HIV-1 model parameters with literature sources.

3.4 S3.4. Monte Carlo Parameter Variability

Inter-individual variability modeled via Latin hypercube sampling:

- $\beta \sim \mathcal{U}(0.7\beta_0, 1.3\beta_0)$
- $\delta \sim \mathcal{U}(0.7\delta_0, 1.3\delta_0)$
- $k \sim \mathcal{U}(0.7k_0, 1.3k_0)$
- $\beta_E \sim \mathcal{U}(0.7\beta_{E0}, 1.3\beta_{E0})$

This yields coefficient of variation $\approx 20\%$ per parameter, matching clinical observations.

4 S4. Influenza A Simulation Details

4.1 S4.1. Baccam Model Equations

Target epithelial cells (T), eclipse phase (E), productive infection (I), virus (V), interferon (F):

$$\dot{T} = -\beta T V, \tag{25}$$
$$\dot{E} = \beta T V - kE, \tag{26}$$
$$\dot{I} = kE - \delta I - mFI, \tag{27}$$
$$\dot{V} = pI - cV - (1 - u(t))\kappa V, \tag{28}$$
$$\dot{F} = qI - \delta_F F, \tag{29}$$

where $\beta = 3.2 \times 10^{-5}$ TCID_{50}^{-1}/day, $k = 4.0$ day^{-1} (eclipse exit), $\delta = 5.2$ day^{-1} (infected cell death), $p = 4.6 \times 10^{-2}$ TCID_{50}/(cell·day), $c = 10$ day^{-1} (viral clearance), $\kappa = 2.0$ day^{-1} (neuraminidase activity), $m = 10^{-5}$ TCID_{50}^{-1}/day (interferon antiviral effect), $q = 10^{-3}$ day^{-1} (interferon production), $\delta_F = 2.0$ day^{-1} (interferon decay).

Drug efficacy $u(t)$ represents neuraminidase inhibitor effectiveness (oseltamivir).

4.2 S4.2. Parameter Calibration

Parameters fitted to Hayden 1999 human challenge data. Initial conditions: $T(0) = 4 \times 10^8$ cells, $E(0) = 0$, $I(0) = 0$, $V(0) = 0.75$ TCID_{50} (viral inoculum), $F(0) = 0$.

5 S5. Decision Tree Implementation and Control Theory

5.1 S5.1. PI Controller Formulation

Proportional-Integral (PI) controller for continuous dose adjustment:

$$u(t) = u_{\text{base}} + K_P(\chi_{\text{target}} - \chi(t)) + K_I \int_0^t (\chi_{\text{target}} - \chi(\tau))d\tau, \tag{30}$$

where K_P is proportional gain, K_I is integral gain. Gains tuned via Ziegler-Nichols method on simulated patient ensemble.

5.2 S5.2. Anti-Windup Implementation

When dose saturates at bounds $[u_{\min}, u_{\max}]$, integral term can accumulate erroneously (windup). Anti-windup uses conditional integration:

$$\frac{dI(t)}{dt} = \begin{cases} \chi_{\text{target}} - \chi(t) & \text{if } u_{\min} < u(t) < u_{\max}, \\ 0 & \text{otherwise.} \end{cases} \tag{31}$$

5.3 S5.3. Sensitivity Analysis: Threshold Variation

Vary decision tree thresholds $\chi_1 \in [0.65, 0.75]$ (underdamped), $\chi_2 \in [0.75, 0.85]$ (caution), $\chi_3 \in [0.95, 1.05]$ (overdamped start), $\chi_4 \in [1.15, 1.25]$ (strongly overdamped).

Results: Qualitative outcomes (toxicity reduction, time in window) robust to ± 0.05 shifts. Optimal performance at $\{\chi_1, \chi_2, \chi_3, \chi_4\} = \{0.70, 0.80, 1.00, 1.20\}$ as stated in main text.

6 S6. Parameter Tables

6.1 S6.1. Complete HIV-1 Parameter Set

Symbol	Value	Units	Description
λ	10^4	cells/(μL·day)	Target cell production
d	0.01	day^{-1}	Target cell death
β	2.4×10^{-5}	μL/(virion·day)	Infection rate
δ	1.0	day^{-1}	Infected cell clearance
k	10^{-4}	μL/(cell·day)	CTL killing rate
p	2000	virions/cell	Viral production
c	23	day^{-1}	Viral clearance
δ_E	0.1	day^{-1}	CTL decay
β_E	0.01	μL/(cell·day)	CTL stimulation
K	1.5×10^3	cells/μL	Carrying capacity

Table 4: Complete HIV-1 model parameter set.

6.2 S6.2. Complete Influenza A Parameter Set

7 S7. COAST Trial: Complete Protocol

7.1 S7.1. Inclusion/Exclusion Criteria

Inclusion:

Symbol	Value	Units	Description
T_0	4×10^8	cells	Initial target cells
β	3.2×10^{-5}	TCID_{50}^{-1}/day	Infection rate
k	4.0	day^{-1}	Eclipse phase exit
δ	5.2	day^{-1}	Infected cell death
p	4.6×10^{-2}	TCID_{50}/(cell·day)	Viral production
c	10	day^{-1}	Viral clearance
κ	2.0	day^{-1}	Neuraminidase activity
m	10^{-5}	TCID_{50}^{-1}/day	Interferon effect
q	10^{-3}	day^{-1}	Interferon production
δ_F	2.0	day^{-1}	Interferon decay

Table 5: Complete influenza A model parameter set.

- HIV-1 infected adults age 18–65
- Treatment-naive (no prior ART)
- CD4 count > 200 cells/μL
- Viral load > 1000 copies/mL
- No active opportunistic infections
- Willing to comply with weekly viral load monitoring

Exclusion:

- Pregnancy or breastfeeding
- Hepatic dysfunction (ALT/AST $> 3\times$ upper limit normal)
- Renal insufficiency (eGFR < 50 mL/min)
- Active substance use disorder
- Prior documented drug resistance

7.2 S7.2. Randomization and Blinding

Randomization 1:1 via permuted blocks (block size 4–8). Stratified by baseline viral load ($< 10^5$ vs $\geq 10^5$ copies/mL) and CD4 count (< 350 vs ≥ 350 cells/μL). Open-label due to different monitoring schedules.

7.3 S7.3. Statistical Analysis Plan

Primary efficacy (non-inferiority):

- Endpoint: Proportion with VL < 50 copies/mL at week 48
- Non-inferiority margin: -10 percentage points
- Test: One-sided 97.5% CI for difference (experimental $-$ control)
- Power: 90% assuming true rates 96% (both arms), $n = 350$ per arm

Primary safety (superiority):

- Endpoint: Incidence of Grade 3–4 adverse events

- Test: Chi-squared test, two-sided $\alpha = 0.05$
- Power: 85% assuming 20% (control) vs 16% (experimental), $n = 350$ per arm

Missing data: Multiple imputation (10 imputations) for missing week 48 viral loads.

7.4 S7.4. Data Safety Monitoring Board

Independent DSMB reviews safety data quarterly. Stopping rules:

- > 3 unexpected serious adverse events attributable to χ-guided strategy
- Conditional power for primary efficacy $< 20\%$ at interim (n=350 enrolled)
- Overwhelming efficacy: 99.9% posterior probability of non-inferiority

8 S8. Cost-Effectiveness Analysis Framework

8.1 S8.1. Incremental Cost Estimation

MTD strategy costs (per patient-year):

- Drug cost: \$15,000 (standard ART regimen)
- Viral load monitoring: 4 tests/year $\times$ \$175 = \$700
- Clinic visits: 4/year $\times$ \$150 = \$600
- Toxicity management: \$2,500 (Grade 3–4 events at 20% incidence)
- **Total: \$18,800/patient-year**

χ-guided strategy costs (per patient-year):

- Drug cost: \$11,000 (reduced dose, 70% of standard)
- Viral load monitoring: 24 tests/year $\times$ \$175 = \$4,200
- Clinic visits: 8/year $\times$ \$150 = \$1,200
- Toxicity management: \$1,500 (16% incidence)
- **Total: \$17,900/patient-year**

Incremental cost: −\$900/patient-year (cost-saving)

8.2 S8.2. QALY Estimation

Quality-adjusted life years gained through reduced toxicity and improved adherence. Utilities (SF-6D):

- Suppressed VL, no toxicity: 0.85
- Suppressed VL, Grade 3–4 toxicity: 0.65
- Unsuppressed VL: 0.50

Expected QALYs over 10 years (3% discounting):

- MTD: 7.2 QALYs

- χ-guided: 7.6 QALYs
- **Incremental: +0.4 QALYs**

Incremental cost-effectiveness ratio (ICER):

$$\text{ICER} = \frac{-\$900/\text{year} \times 8.5 \text{ years}}{0.4 \text{ QALYs}} = -\$19,125/\text{QALY}. \tag{32}$$

Negative ICER indicates χ-guided strategy is dominant (cost-saving and health-improving).

9 S9. Figures

9.1 S9.1. Sensitivity Heatmaps

(Placeholder for heatmaps showing virologic success rate and cumulative toxicity as functions of parameter pairs (r, α), (δ, β), (k_u, ω).)

9.2 S9.2. Individual Patient Trajectories

(Placeholder for 10 representative individual patient viral load and χ trajectories from Monte Carlo ensemble.)

9.3 S9.3. Dose Adjustment Histories

(Placeholder for histogram of dose changes per patient under χ-adaptive protocol.)